

```
1  # Name: Christopher A. Lee
2  # Date: 02/28/2025
3  # Prof: Dr. Callura
4  # Course: Bioinformatics – ENBC 455
5
6  from Bio import SeqIO
7  from Bio import Align
8  from Bio import SeqUtils
9  # Problem 1
10 with open('hw5_1.txt', 'r') as f:
11     seq1 = [str(fasta_seq.seq) for fasta_seq in SeqIO.parse(f, "fasta")]
12
13     aligner1 = Align.PairwiseAligner()
14     aligner1.substitution_matrix = Align.substitution_matrices.load("BLOSUM62")
15     aligner1.open_gap_score = -5
16     aligner1.extend_gap_score = -5
17
18     max_score = aligner1.score(seq1[0], seq1[1])
19
20     print(f"Problem 1\nMaximum Alignment Score: {max_score}\n")
21
22 # Problem 2
23 with open('hw5_2.txt', 'r') as f:
24     seq2 = [str(fasta_seq.seq) for fasta_seq in SeqIO.parse(f, "fasta")]
25
26     aligner2 = Align.PairwiseAligner()
27     aligner2.substitution_matrix = Align.substitution_matrices.load("PAM250")
28     aligner2.mode = 'local'
29     aligner2.open_gap_score = -5
30     aligner2.extend_gap_score = -5
31
32     alignments = aligner2.align(seq2[0], seq2[1])
33     print(f"Problem 2\n{alignments[0].score}\n{alignments[0][0].replace('-', ' ')}\n{alignments[0][1].replace('-', ' ')}\n")
34
35 with open('hw5_4.txt', 'r') as f:
36     seq3 = [fasta_seq for fasta_seq in SeqIO.parse(f, "fasta")]
37     seqs = [str(fasta_seq.seq) for fasta_seq in seq3]
38
39     most_gc_content = 0
40     most_gc_content_index = 0
41     for i, seq in enumerate(seqs):
42         temp = SeqUtils.gc_fraction(seq)
43         if temp > most_gc_content:
44             most_gc_content = temp
45             most_gc_content_index = i
46
47     print("Problem 4")
48     print(seq3[most_gc_content_index].id)
49     print(f"{most_gc_content * 100:.6f}%")
```


Problem 1
Maximum Alignment Score: 2045.0

Problem 2
1476.0

NETALCIPFSMRSIQQGD LGKQLDLAKCFYGKRQSDGDVGT VNTINICWFGLHLHYYYIGCLISQMSDKCSRTWKRVIKGSCNIQHIVHIAAYLCDEF LDMFTERLMGAWK IACESWIW MYQWSVRCDATTWWEWGWYIERFEIHRCTGECGRICNSCFWPENTFFVTLGTQSNG
YFSVIGQWIHRHNYKLCPQRNYCFRWYPLYKQDRMRIRDL DGQGRQRQWNKQIKSCMMKEEDRIPC METRSWCYSLNWVAPLKVWYQLYHVRHCQHRTMTYYDDTFFGFHWWEWPCGHQSLKLAL THSQIAQTICWLVDHHADKRPEGDCVDSGGGMWVCQQMGAPIHDNTNF
RTGMIWCSWG TKEYFAHPTQSRHACRDIWQRRLHHRDNCWKYMKICIGTFQSMWPAINCGNDGSPLDQWYILIGVCGGYMWINWDGFAPPQIETLATGANNCVKRPNWVQCKSWTAIHQALFYVYCEMFEC DPTFDGIDEAIMLQACPQALRIAECRPKFVPPICHWIFSVLRPP
SNMRYTCYKTS L FVNYDTADMAHNEEEWYMDVRCIPACLKMCLAMLMQYAVGRWIMEWPCWTFYFNCCCLNPAENFCRIWHNMDFWSIEAPWFGIDGRIGDGQWDHIKVCDDNFYCQTPENHMPHQPLCARWVHYGAMIYHGTYQRLKLYHWHPNQTEIWIDENQPHETMDCK
RTSHCMQDYCSMGGILWDAQTDNFEHMNTWLWFAPFVMFWMAECQNGHSYDSISWQYAEQWSRHLKWKDAHNGYCQIYIYHCHNIYDLPWNMGFMHGNAMHKDRWAF CQQVGDSTEHIREDFVIINCDHEDYLNEFRLCATDQMHEHAHGYPMPMCALRYFDGTVFSDCGPCEGQ
YQ
HRTEACCRFKWMRLNYAMGLHGQCLIVYRNTNMLS NFWKYADWKCYNYIAFMLPPARIVCDSSVNYNTLARKFAIAIRSRHNIWMHHEWQWNSREMTLDIDQNIQARHWCIAHGFYLLFAHVCWADYLGCCMCYPWWQDQDEFMLDFFVHRHQPYAEPCTSFMAPGYFPLDTSRC
RMCYEQSSLENFCTPOHDPNDIPC MQKTPATSWNRNKENNRNWHEKKWQEIAASQTLN YEQQD VWPRQROWNIQIKSCMMKNDDRIPC METRPWCYSLNWVAVLTNIEALKVQLYHVRHCQKRTMTYIDDTFKGFHLWCSKENPCGIFEKGEDNQOETSQVQIAQTICWLVDHAA
DKRPEGESMAGCVESGGGMWVCQAYAMGAPIHDNTPEETLAILAAMAWCKWG TKEYFQMI ECQKRHPTASMHACRDIWQRALHHRDNC PKGGDQKTKYMKICIGTFQRMWKYPHRYIADGSPGDGLGWYILIGVCGGYMWINWDGFAPPQKIEYVYVWNCVKRPNWNWTAITQA
LFYFCPTFDGEENPIASMLQALWLAECRIHKRTKFVSPICHWIVLRPPSNMTSLFVIHHGHINYGTAD EPDTFKVR CIPACLNPTVPYSVGRWIMWPNWTFYFNCCCLVMIMLP IKVGRGIKEVMRPTMIANWWWPDQWGPILMACQOKPQYRINA AVLRTFSQFKAPHQWW
FMYIKYRK FQPV IYGRGIVDNVCEEAMSKHSIRODDAIGKKMKAGFLRKPSNPSNGLPPAPCYCSNANMACFIGPEQHSQPTCITDTVKHDHVRQFP IQCWMAKR NKDLSCKDMGMFHAMPEGQAYFMQLANEDLVAKIFKHRQHPICLSLPFWVLRWLMVNISGMAGIHEAFMP
NSKL GKWGVHYEANEPM EKLRILGVMIQHFVKA IAGKFK

Problem 4
Rosalind_3475
53.680634%

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Multiple Sequence Alignment (MSA)

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CLUSTAL O(1.2.4) multiple sequence alignment

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Rosalind_4982	AAAGGTGT---	8
Rosalind_1754	AACTGGGCTG-	10
Rosalind_1893	---TGTGCATT	8
Rosalind_9676	---GAGTCTAG	8

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